

Question ID: eae66bf9

Question

Text 1

In 2021, a team led by Amir Siraj hypothesized that the Chicxulub impactor—the object that struck the Yucatán Peninsula sixty-six million years ago, precipitating the mass extinction of the dinosaurs—was likely a member of the class of long-period comets. As evidence, Siraj cited the carbonaceous chondritic composition of samples from the Chicxulub impact crater as well as of samples obtained from long-period comet Wild 2 in 2006.

Text 2

Although long-period comets contain carbonaceous chondrites, asteroids are similarly rich in these materials. Furthermore, some asteroids are rich in iridium, as Natalia Artemieva points out, whereas long-period comets are not. Given the prevalence of iridium at the crater and, more broadly, in geological layers deposited worldwide following the impact, Artemieva argues that an asteroid is a more plausible candidate for the Chicxulub impactor.

Based on the texts, how would Artemieva likely respond to Siraj's hypothesis, as presented in Text 1?

Answer

- A. By insisting that it overestimates how representative Wild 2 is of long-period comets as a class
- B. By arguing that it does not account for the amount of iridium found in geological layers dating to the Chicxulub impact
- C. By praising it for connecting the composition of Chicxulub crater samples to the composition of certain asteroids
- D. By concurring that carbonaceous chondrites are prevalent in soil samples from sites distant from the Chicxulub crater